Anruo Zhong

PhD student in physics with a background in energy engineering.

My keywords: artificial intelligence, computational materials science and mechanics, scientific software development.

2 April 1998

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anruo-zhong

zhonganr

Personal Qualities

Fast learner, critical thinking, eye for detail, creative, organized.

Human languages

- English (C1, TOEFL: 104/120)
- French (B2-C1)
- Chinese (Native)
- Cantonese (Native)

Programming languages

Python
C/C++
Fortran
Matlab
SQL
JavaScript

Professional skills

- Numerical simulations: LAMMPS, VASP, ANSYS (Mechanical, FLUENT, ICEM), HyperMesh, SolidWorks, CATIA.
- Programming and computing tools: Open MPI, oneAPI MKL (LAPACK/ScaLAPACK), HPC, Git, OpenGL, Docker, shell script.
- Machine learning: Scikit-Learn, TensorFlow, JAX, Apache TVM.

EXPERTISE

Materials Science
- Behavior of metallic materials under extreme conditions
- Thermodynamics, mechanics and phase diagrams of solid materials
- Microstructural analysis (from simulations and experiments)

Data science - Machine Learning and Deep Learning
- Predictive models: design, optimization and reliability analysis
- Data analysis: data visualization, anomaly detection, classification and clustering
- Database: design, construction and management

WORK EXPERIENCE

PhD student at CEA, SRMP, Gif-sur-Yvette, France 10/2021 – 10/2024
- Contribution to MILADY (Machine Learning Dynamics) package: implementation of machine learning force fields for molecular dynamics (MD). Design and building of databases. https://ai-atoms.github.io/milady-docs
- Contribution to MILADY-LAMMPS: an extended development of the community MD software LAMMPS to make it compatible with the MILADY-constructed force fields.
- Development of Bayesian adaptive biasing force method for enhanced free energy sampling and its applications: data-driven prediction of thermodynamic, mechanical properties and defects behaviors for metallic materials.

AI Engineer Intern at OPEN AI Lab, Shenzhen, China 06/2021 – 09/2021
- Feasibility study of ML compiler framework Apache TVM: pass infrastructure and quantization.

Research Intern at Chinese Academy of Sciences, Guangzhou, China 11/2020 – 05/2021
- Development of micromagnetic simulation software: multi-geometry meshing, parallel PDE solvers, GUI and interface to FEM tool (Ansys Mechanical). https://github.com/zhonganr/NMD
- Numerical simulation of magnetoelastic interaction in chiral magnets for potential data storage devices: mechanical manipulation of skyrmions in racetrack memory.

Mechanical Engineer Intern at China General Nuclear Power Engineering Company Ltd., Shenzhen, China 07/2020 – 09/2020
- Finite element analysis (Ansys Mechanical) of nuclear island buildings under accident conditions and participation in report writing for Generic Design Assessment in the UK.

Software Engineer Intern at OPEN AI Lab, Shenzhen, China 07/2019 – 08/2019
- Implementation of accelerated matrix multiplication algorithm.

Operator Intern at Yangjiang Nuclear Power Station, Yangjiang, China 07/2018 - 08/2018
- Daily inspection and maintenance of electrical equipment.

EDUCATION

Université Paris-Saclay, Gif-sur-Yvette, France 10/2021 – 10/2024
- Degree: PhD in Physics (supervisors: Mihai-Cosmin Marinica, Manuel Athènes)
- Thesis: Predicting temperature effects on thermodynamics of alloy via high-dimensional regression metamodels

Sun Yat-sen University, Guangzhou, China (globally ranked #72 by Shanghai Ranking) 08/2019 – 06/2021
- Degree: Master of Engineering in Nuclear Engineering (grade: 87.8/100)
- Specialization: computational mechanics and reactor physics
- Thesis: Dynamics and stability of Bloch skyrmions in inhomogeneous elastic fields

Grenoble INP - Ense3, Grenoble, France 12/2019 – 06/2020
- Exchange semester (grade: 16.8/20, *mention très bien*)

Sun Yat-sen University, Guangzhou, China 08/2015 – 06/2019
- Degree: Bachelor of Engineering in Nuclear Engineering (grade: 3.4/4)
- Thesis: Molecular dynamics simulations of radiation damage in Ni-based alloys

National Undergraduate Admission Exam, China 06/2015
- Option science, ranked top 0.8% (grade: 646/750)

Other interests

Cycling, hiking, badminton, piano (national grade: 8/10), traveling, bakery.

References

Mihai-Cosmin Marinica
Senior researcher, CEA Saclay
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Manuel Athènes
Senior researcher, CEA Saclay
manuel.athenes@cea.fr

Jean-Luc Béchade
Head of laboratory, CEA Saclay
jean-luc.bechade@cea.fr

AWARDS

Sun Yat-sen University Excellent Student Scholarship (6 consecutive years)	2015 – 2021
CGN Excellent Graduate Student Scholarship	12/2019
Sun Yat-sen University Excellent Bachelor Program & Best Poster Award	06/2019
CentraleSupélec Jury Excellence Prize for Engineering Degree Program	10/2018

PUBLICATIONS

- Wan, X., **Zhong, A.**, *et al.* (2024). Discontinuous to continuous transition changeover and magnetic helicity reversal in helimagnet nanodisks under torsion. *New J. Phys.* 26 023009.

- **Zhong, A.**, Lapointe, C., *et al.* (2023). Anharmonic thermo-elasticity of tungsten from accelerated Bayesian adaptive biasing force calculations with data-driven force fields. *Phys. Rev. Mater.* 7 023802.

- **Zhong, A.**, Lan, X., *et al.* (2022). Dynamics and stability of skyrmions in a bent nano-beam. *New J. Phys.* 24 033019.